A/B Testing Comparison using Data Analytics

Project Name: A/B Testing Comparison of Facebook Ads and Google AdWords

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1. Introduction

Digital advertising has become one of the most powerful marketing strategies used by organizations to reach potential customers. Businesses spend significant budgets on platforms such as Facebook Ads and Google AdWords to increase brand awareness, website traffic, and conversions. However, selecting the most effective advertising platform is an important business decision.

This project performs an A/B Testing Comparison between Facebook Ads and Google AdWords campaigns. The analysis focuses on evaluating advertisement performance using statistical techniques and data analytics. Different performance metrics including ad views, clicks, conversions, click-through rate (CTR), conversion rate, and cost per click (CPC) are analyzed to determine which platform provides better marketing outcomes.

2. Project Objectives

The primary objectives of this project are:

- Compare Facebook Ads and Google AdWords campaign performance.
- Analyze advertisement clicks and conversions.
- Evaluate campaign cost effectiveness.
- Perform statistical hypothesis testing.
- Study relationships between clicks and conversions.
- Predict conversions using Linear Regression.
- Generate business insights for future advertising strategies.

3. Business Problem

In today's highly competitive digital marketing environment, businesses invest substantial budgets in online advertising to increase brand awareness, attract potential customers, and improve sales. Among the most popular advertising platforms, **Facebook Ads** and **Google AdWords** offer powerful targeting capabilities, but each platform performs differently depending on audience behavior, campaign objectives, and advertising strategies.

For marketing agencies and businesses, choosing the most effective platform is a critical decision because it directly impacts the **Return on Investment (ROI)**, customer acquisition cost, and overall campaign success. Without proper analysis, organizations may allocate large portions of their advertising budget to a platform that does not deliver the best results, leading to unnecessary expenses and lower profitability.

The primary challenge is to determine which advertising platform generates higher conversions while maintaining better cost efficiency. This requires analyzing key performance metrics such as advertisement views, clicks, conversion rates, cost per click (CPC), and overall campaign effectiveness. In addition to comparing campaign performance, it is also important to understand whether the observed differences between the two platforms are statistically significant or simply due to random variation.

This project addresses these challenges by applying **A/B Testing**, **Exploratory Data Analysis (EDA)**, **Hypothesis Testing**, **Correlation Analysis**, **Linear Regression**, and **Time Series Analysis** on historical marketing campaign data. The insights generated from this analysis help marketing teams make data-driven decisions, optimize advertising budgets, improve campaign performance, and maximize overall business profitability.

4. Research Question

This project attempts to answer the following questions:

- Which advertising platform generates more conversions?
- Does increasing advertisement clicks increase conversions?
- Is there a statistically significant difference between Facebook and Google AdWords campaigns?
- Which platform is more cost effective?
- How do conversions vary across different months and weekdays?

5. Dataset Description

The dataset contains marketing campaign data collected from Facebook Ads and Google AdWords campaigns.

The dataset includes information such as:

- Date
- Facebook Ad Views
- Facebook Ad Clicks
- Facebook Ad Conversions
- Facebook Cost
- Facebook Click Through Rate
- Facebook Conversion Rate
- Facebook Cost Per Click
- Google AdWords Views
- Google AdWords Clicks
- Google AdWords Conversions
- Google AdWords Cost
- Google AdWords Click Through Rate
- Google AdWords Cost Per Click

The dataset contains approximately **100,000 records**, making it suitable for performing statistical analysis and predictive modeling.

6. Tools and Technologies

The project was implemented using the following tools and Python libraries:

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy
- Statsmodels
- Scikit-learn

7. Project Methodology

The project followed the following workflow:

Step 1 – Data Collection:

The marketing campaign dataset was loaded into Python using the Pandas library.

Step 2 – Data Cleaning:

The dataset was inspected for data types and converted into suitable formats. Date columns were converted into datetime format and numerical calculations were prepared for further analysis.

Step 3 – Exploratory Data Analysis:

EDA was performed to understand campaign performance through descriptive statistics and visualizations.

Step 4 – Data Visualization:

Several visualizations were created including:

- Histograms
- Scatter Plots
- Bar Charts
- Line Charts

These graphs helped identify advertising trends and campaign behavior.

Step 5 – Correlation Analysis

Correlation analysis was performed to understand whether advertisement clicks influence conversions.

Step 6 – Hypothesis Testing

Independent Two Sample T-Test was performed to determine whether Facebook Ads generate significantly more conversions than Google AdWords.

Step 7 – Regression Analysis

Linear Regression was used to predict expected conversions based on advertisement clicks.

Step 8 – Time Series Analysis

Monthly and weekly conversion trends were analyzed to understand seasonal behavior.

Step 9 – Cost Per Conversion Analysis

Cost efficiency was analyzed by calculating monthly Cost Per Conversion values.

Step 10 – Cointegration Test

Long-term equilibrium between advertising cost and conversions was examined using Cointegration Analysis.

8. Exploratory Data Analysis (EDA)

The following analyses were performed:

- Dataset overview
- Descriptive statistics
- Distribution of clicks
- Distribution of conversions
- Conversion frequency analysis
- Monthly conversion trends
- Weekly conversion trends

The visualizations helped identify campaign performance patterns and customer engagement behavior.

9. Statistical Analysis

An Independent Two Sample T-Test was conducted.

1) Null Hypothesis (H_0)

There is no significant difference between Facebook Ads and Google AdWords conversions.

2) Alternative Hypothesis (H_1)

Facebook Ads generate significantly higher conversions than Google AdWords.

The obtained p-value was greater than the significance level (0.05), indicating that the difference between the two platforms is not statistically significant.

Therefore, the null hypothesis was accepted.

```
Mean Conversion
 --------------
Facebook: 150.53
AdWords: 150.84

T statistic -0.5190299450859179
p-value 0.6037404336166815

p-value is less than significance value, Fail to reject the null hypothesis
```

10. Regression Analysis

A Linear Regression model was developed using Facebook Advertisement Clicks as the independent variable and Facebook Advertisement Conversions as the dependent variable.

The regression model demonstrated a moderate positive relationship between clicks and conversions.

The model can be used to estimate expected conversions based on future advertisement clicks, providing useful insights for campaign planning and budget allocation.

```
Accuracy(R2 Score): 42.4
Mean Squared Error:  9987.22
```

```
For 50 clicks, Expected Conversion : 13.23 conversions
For 80 clicks, Expected Conversion : 20.69 conversions
```

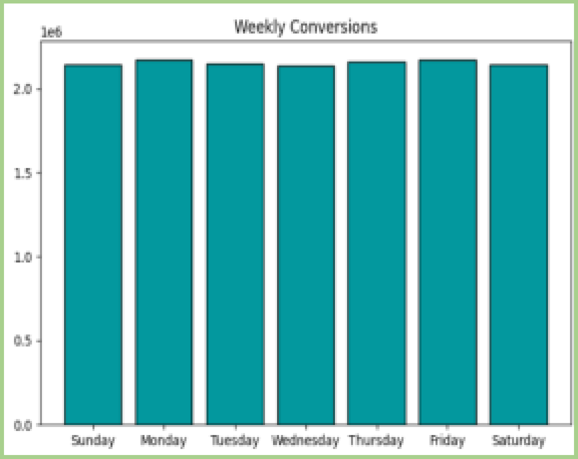
11. Time Series Analysis

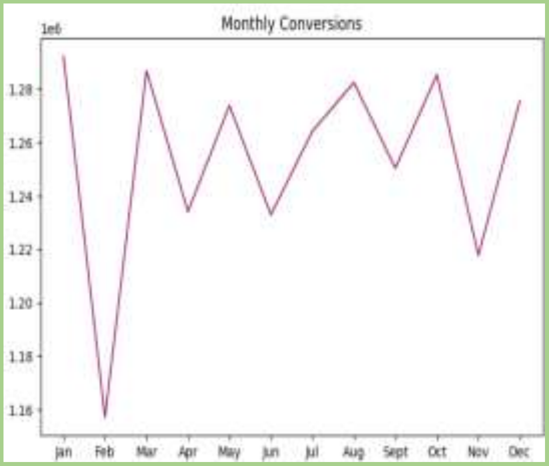
Time series analysis was conducted to identify monthly and weekly conversion trends.

The analysis showed relatively stable conversion performance throughout the year with moderate seasonal fluctuations.

September recorded the lowest Cost Per Conversion, suggesting improved advertising efficiency during that period.

These insights can assist businesses in optimizing advertising budgets according to seasonal performance.





12. Results and Findings

The project produced the following major findings:

- Facebook and Google AdWords generated similar average conversion performance.
- Advertisement clicks showed a positive relationship with conversions.
- The regression model successfully predicted expected conversions.
- Monthly conversion trends indicated seasonal fluctuations.
- September recorded the lowest Cost Per Conversion.
- Statistical testing showed no significant difference between Facebook and Google AdWords conversions.
- Both platforms demonstrated effective advertising performance.

13. Conclusion

This project successfully compared Facebook Ads and Google AdWords advertising campaigns using statistical analysis and machine learning techniques.

The analysis indicates that both advertising platforms perform similarly with respect to average conversions. Rather than relying solely on platform selection, organizations should focus on audience targeting, campaign optimization, advertisement quality, and budget management.

The developed analytical workflow can help marketing agencies improve advertising strategies and make informed business decisions based on data.

14. Future Scope

The project can be extended by incorporating:

- Machine Learning Classification Models
- Customer Segmentation
- Real-Time Interactive Dashboards
- ROI Prediction Models
- Deep Learning Algorithms
- Marketing Automation Systems
- Predictive Analytics
- Cloud-Based Data Processing

15. References

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